MIL-B-85667/2A
28 January 1991
SUPERSEDING
MIL-B-85667/2(AS)
14 March 1986

MILITARY SPECIFICATION SHEET

BOLT, SLEEVE, 100° REDUCED FLUSH HEAD, STRAIGHT SHANK, CRES 304 SLEEVE, CRES PH 13-8Mo PIN, 125 KSI Fsu

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-85667.

1. Application criteria. The sleeve bolt shall be used to provide a precision net fit or controlled interference fit. The sleeve bolts shall be used in accordance with the limitations of MIL-B-85667/1(AS). The sleeve bolts shall not be degreased and cetyl alcohol shall not be removed before installation.

2. Companion and accessory parts. Nuts in accordance with MIL-N-85730 shall be used with these sleeve bolts.

3. Oversize bolts First and second oversize sleeve bolts may replace MIL-B-85667/2 sleeve bolts for first and second repair operations.

4. Installation. Installation shall be in accordance with MIL-HDBK-271-104.

5. Materials.

5.1 Pin. Pin material shall be CRES PH 13-8Mo (UNS S13800) age-hardenable steel in accordance with AMS 5629.

5.2 Sleeve. Sleeve material shall be AISI 304 (UNS S30400) in accordance with MIL-T-8504.

6. Heat treatment.

6.1 Pin. Solution treat in air, vacuum or inert gas atmosphere at 1700 ±25°F for one hour. Air or oil cool; when solution treating in vacuum or inert gas atmosphere, inert gas or nitrogen may be used for cooling Material shall be cooled at -100°F for two hours within 24 hours after solution treatment. Precipitation harden 4 hours minimum at the highest temperature (950-1000°F) necessary to achieve the required strength. No subsequent stress relieving is permitted. However, low temperature embrittlement relief treatments may be used if applicable.

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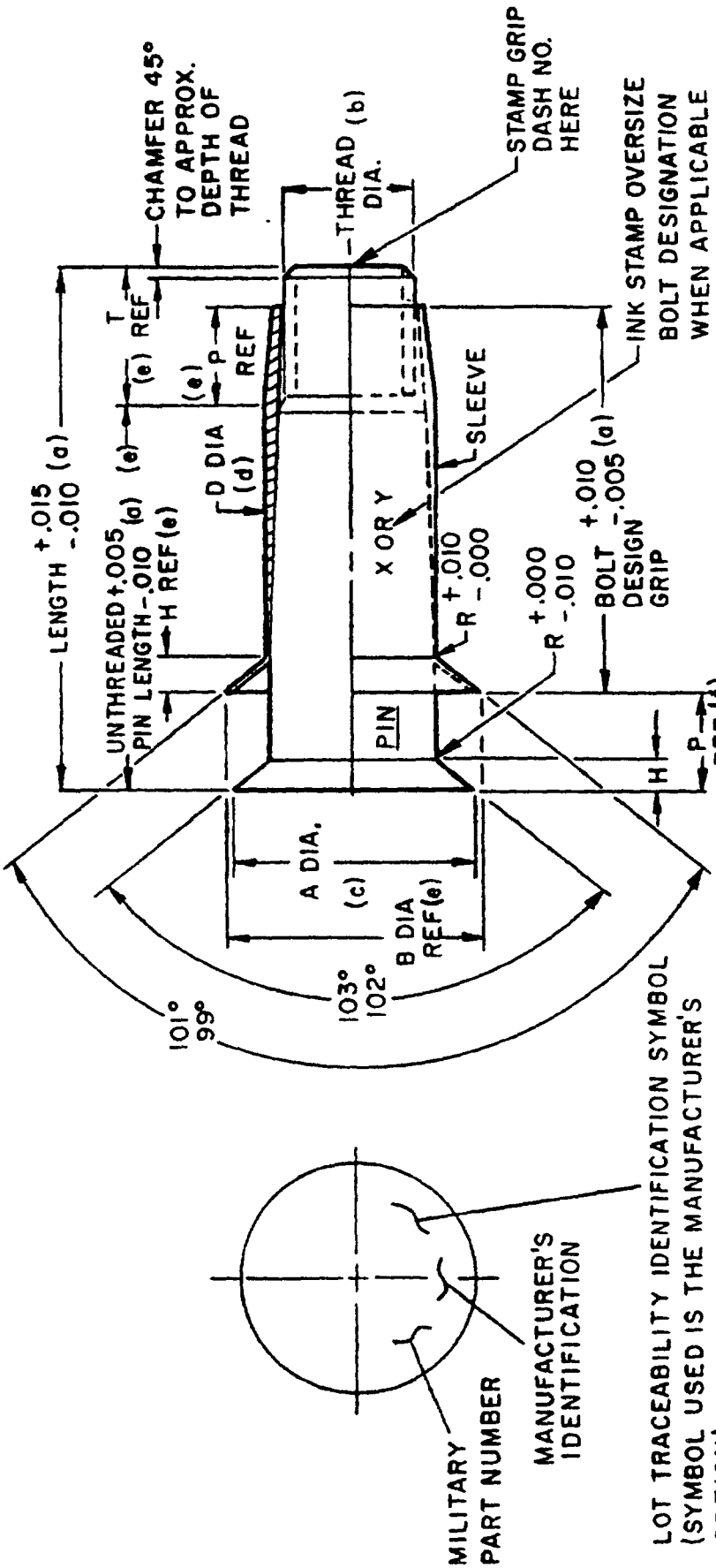


FIGURE 1. Reduced flush head bolt.

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NOTES:

a. Grip lengths are available in .0625 increments.

b. Threads shall be in accordance with MIL-S-8879. Thread major diameter (T.D.) shall be in accordance with Table I.

c. 1. For sizes .190 thru .375, dimension "A" shall be concentric to dimension "D" within .008 FIM.

2. For sizes .437 thru .625, .012 FIM.

3. For sizes .750 thru 1.000, .020 FIM.

d. 1. For sizes .190 thru .375, dimension "D" shall be concentric to thread pitch diameter within .006 FIM.

2. For sizes .437 thru 1.000, .010 FIM.

e. Reference dimensions are for design purposes only and are not an inspection or manufacturing requirement.

f. Protrusion will vary according to taper to provide expansion as given in Table I.

6.1.1 Hardness. Pin hardness shall not be greater than 48 HRC.

6.2 Sleeve. Sleeve is cold worked.

7. Surface texture. Surface texture shall be R_a-32 in accordance with ANSI/ASME B46.1.

8. Coatings or surface material.

8.1 Pin. The pin shall be dry-film lubricated in accordance with MIL-L-46010, Type I, .0003/.0005 thick.

8.2 Sleeve. None.

9. Lubricants. 0.0001-inch overspray allowed on head and threads. In addition, the pin outside diameter and sleeve inside diameter shall be lubricated with cetyl alcohol in accordance with MIL-L-87132.

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TABLE I. Bolt dimensions.

SIZE NO'S	GROUP	GRIP DASH NUMBERS	THREAD PER MIL-S-8879 UNJF-3A	A BOLT HEAD DIA	B REF SLEEVE FLANGE DIAMETER			N	FH FLANGE HEIGHT REF	P (f) REF	R FILLET RADIUS	T D THREAD DIA	T THREAD LENGTH	D DIMENSION INITIAL SLEEVE DIAMETER (a)			D' DIMENSION FINAL SLEEVE DIAMETER (a) (b)			MINIMUM INSTALLATION HOLE SIZE DIAMETER		
					STD	X O/S	Y O/S							STD	X O/S	Y O/S	STD	X O/S	Y O/S	STD.	X O/S	Y O/S
-3	1	-6 THRU -8	1900-32	.3327 .3259	.3820	.3880	.3940	.048 .046	.061	.225 .215	.030	.1840 .1810	.343	.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	2	-9 THRU -16		.3341 .3273	.3838	.3898	.3958		.062	.432 .422				.2365 .2360	.2425 .2420	.2485 .2480	.2455 .2445	.2515 .2505	.2575 .2565	.237	.243	.249
	3	-17 THRU -24		.3445 .3377	.4004	.4074	.4144		.064	.432 .422				.2485 .2480	.2555 .2550	.2625 .2620	.2575 .2565	.2645 .2635	.2715 .2705	.249	.256	.263
-4	1	-6 THRU -8	2500-28	.4295 .4224	.4772	.4882	.4942	.063 .061	.074	.225 .215	.030	.2440 .2410	.406	.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	2	-9 THRU -17		.4339 .4268	.4805	.4915	.4975		.075	.432 .422				.3005 .3000	.3115 .3110	.3175 .3170	.3095 .3085	.3205 .3195	.3265 .3255	.301	.312	.318
	3	-18 THRU -28		.4469 .4398	.5011	.5081	.5171		.078	.432 .422				.3145 .3140	.3215 .3210	.3305 .3300	.3235 .3225	.3305 .3295	.3395 .3385	.315	.322	.331
-5	1	-6 THRU -8	.3125-24	.5080 .5013	.5556	.5626	.5696	.070 .068	.080	.225 .215	.040	.3060 .3020	.484	.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	2	-9 THRU -20		.5163 .5090	.5630	.5700	.5770		.082	.432 .422				.3665 .3660	.3735 .3730	.3805 .3800	.3755 .3745	.3825 .3815	.3895 .3885	.367	.374	.381
	3	-21 THRU -31		.5306 .5233	.5836	.5916	.5986		.086	.432 .422				.3805 .3800	.3885 .3880	.3955 .3950	.3895 .3885	.3975 .3965	.4045 .4035	.381	.389	.396
-6	1	-6 THRU -8	3750-24	.5977 .5876	.6434	.6514	.6634	.081 .078	.090	.225 .215	.040	.3680 .3640	.562	.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	2	-9 THRU -22		.6087 .5986	.6536	.6616	.6736		.093	.432 .422				.4315 .4310	.4395 .4390	.4515 .4510	.4405 .4395	.4485 .4475	.4605 .4595	.432	.440	.452
	3	-23 THRU -36		.6243 .6142	.6754	.6834	.6914		.096	.432 .422				.4475 .4470	.4555 .4550	.4635 .4630	.4565 .4555	.4645 .4635	.4725 .4715	.448	.456	.464
-7	1	-6 THRU -8	4375-20	.7072 .6968	.7571	.7691	.7771	.100 .097	.108	.225 .215	.050	.4310 .4260	.625	.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5165	.5275 .5265	.499	.511	.519
	2	-9 THRU -26		.7234 .7130	.7729	.7849	.7929		.115	.432 .422				.4985 .4980	.5105 .5100	.5185 .5180	.5075 .5065	.5195 .5185	.5275 .5265	.499	.511	.519
	3	-27 THRU -40		.7416 .7312	.7974	.8084	.8204		.117	.432 .422				.5185 .5180	.5295 .5290	.5415 .5410	.5275 .5265	.5385 .5375	.5505 .5495	.519	.530	.542
-8	1	-6 THRU -8	5000-20	.7937 .7831	.8425	.8525	.8625	.110 .107	.114	.225 .215	.050	.4930 .4880	.703	.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	2	-9 THRU -30		.8157 .8051	.8637	.8737	.8837		.123	.432 .422				.5695 .5690	.5795 .5790	.5895 .5890	.5785 .5775	.5885 .5875	.5985 .5975	.570	.580	.590
	3	-31 THRU -46		.8366 .8260	.8908	.9008	.9098		.127	.432 .422				.5895 .5890	.5995 .5990	.6085 .6080	.5985 .5975	.6085 .6075	.6175 .6165	.590	.600	.609

NOTES

(a) D" DIA SHALL BE VERIFIED WITHIN ITS GRIP BEARING PORTION BEFORE EXPANSION WHEN SLEEVE IS PUSHED TO BOLT HEAD, THE D DIA SHALL BE WITHIN THE EXPANDED "D" DIA LIMITS

(b) THIS BOLT IS MANUFACTURED AS A PRECISION ASSEMBLY REMOVAL OF SLEEVE FROM PIN BY USER WILL RESULT IN LOSS OF THE BOLT'S ABILITY TO MEET MIN/MAX EXPANSION LIMITS.

(c) ALL DIMENSIONS ARE IN INCHES TOLERANCES DECIMALS ± 010, ANGLES ±1°

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TABLE II. Mechanical properties.

Diameter Dash Number	Double Shear Strength Lbs-Min (a)	Pin Test Diameter ±.0005	Pin Test Dia Area (a) In.^{2}	Max Install Torque Inch-Lbs	Ultimate Tensile Strength Lbs-Min
-3	7,090	.1900	.0284	40	2,700
-4	12,300	.2500	.0491	85	4,700
-5	19,200	.3125	.0767	140	7,400
-6	27,600	.3750	.1104	200	10,500
-7	37,500	.4375	.1503	300	14,500
-8	49,200	.5000	.1963	455	19,000

Notes:

a. Shear test performed on pins ground to pin test diameter. The shear values are based on a minimum shear strength of 125 KSI Fsu and the stress area at nominal test diameter.

b. Tensile fatigue and stress durability tests not applicable to 100° reduced flush head configuration.

10. Example of military part number.

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- 16 — Grip length dash number in sixteenths of an inch. (Stamp on threaded end.)
- - — '-' designates basic diameter: X designates 1st oversize, Y designates 2nd oversize. (Ink stamp on sleeve when applicable.)
- 3 — Diameter dash numbers in sixteenths of an inch.
- M85667/2 — Basic specification sheet number.

11. Administrative and contractual provisions.

a. For design purposes and in case of conflict, this document takes precedence over documents referenced herein.

b. The bolt assemblies identified herein are proprietary products and are covered by the following patents: U.S. 4,087,896 (expiring May 9, 1997); 4,106,036 (expiring July 25, 1997); 4,048,898 (expiring September 20, 1996); Great Britain 1,524,886; Canada 1,023,585; Japan 50-125256; Italy 1,051,593; Israel 515578; and Germany 2,546,641.5. The Government does not have a royalty-free license.

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TABLE III. Bolt grips and lengths.

Grip Number	Bolt Design Grip	-3 Diameter		-4 Diameter		-5 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.405	.748	.405	.811	.405	.889
7	.438	.468	.811	.468	.874	.468	.952
8	.500	.530	.873	.530	.936	.530	1.014
9	.562	.592	.935	.592	.998	.592	1.076
10	.625	.655	.998	.655	1.061	.655	1.139
11	.688	.718	1.061	.718	1.124	.718	1.202
12	.750	.780	1.123	.780	1.186	.780	1.264
13	.812	.842	1.185	.842	1.248	.842	1.326
14	.875	.905	1.248	.905	1.311	.905	1.389
15	.938	.968	1.311	.968	1.374	.968	1.452
16	1.000	1.030	1.373	1.030	1.436	1.030	1.514
17	1.062	1.092	1.435	1.092	1.498	1.092	1.576
18	1.125	1.155	1.498	1.155	1.561	1.155	1.639
19	1.188	1.218	1.561	1.218	1.624	1.218	1.702
20	1.250	1.280	1.623	1.280	1.686	1.280	1.764
21	1.312	1.234	1.685	1.234	1.748	1.342	1.826
22	1.375	1.405	1.748	1.405	1.811	1.405	1.889
23	1.438	1.468	1.811	1.468	1.874	1.468	1.952
24	1.500	1.530	1.873	1.530	1.936	1.530	2.014
25	1.562			1.592	1.998	1.592	2.076
26	1.625			1.655	2.061	1.655	2.139
27	1.688			1.718	2.124	1.718	2.702
28	1.750			1.780	2.186	1.780	2.264
29	1.812					1.842	2.326
30	1.875					1.905	2.389
31	1.938					1.968	2.452

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TABLE III. Bolt grips and lengths (continued).

Grip Number	Bolt Design Grip	-6 Diameter		-7 Diameter		-8 Diameter	
		Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length	Unthreaded Pin Length (See Figure 1)	Bolt Length
6	.375	.405	.968	.405	1.030	.405	1.108
7	.438	.468	1.030	.468	1.093	.468	1.171
8	.500	.530	1.092	.530	1.155	.530	1.233
9	.562	.592	1.154	.592	1.217	.592	1.295
10	.625	.655	1.217	.655	1.280	.655	1.358
11	.688	.718	1.280	.718	1.343	.718	1.421
12	.750	.780	1.342	.780	1.405	.780	1.483
13	.812	.842	1.404	.842	1.467	.842	1.545
14	.875	.905	1.467	.905	1.530	.905	1.608
15	.938	.968	1.530	.968	1.593	.968	1.671
16	1.000	1.030	1.592	1.030	1.655	1.030	1.733
17	1.062	1.092	1.654	1.092	1.717	1.092	1.795
18	1.125	1.155	1.717	1.155	1.780	1.155	1.858
19	1.188	1.218	1.780	1.218	1.843	1.218	1.921
20	1.250	1.280	1.842	1.280	1.905	1.280	1.983
21	1.312	1.342	1.904	1.342	1.967	1.342	2.045
22	1.375	1.405	1.967	1.405	2.030	1.405	2.108
23	1.438	1.468	2.030	1.468	2.093	1.468	2.171
24	1.500	1.530	2.092	1.530	2.155	1.530	2.233
25	1.562	1.592	2.154	1.592	2.217	1.592	2.295
26	1.625	1.655	2.217	1.655	2.280	1.655	2.358
27	1.688	1.718	2.280	1.718	2.343	1.718	2.421
28	1.750	1.780	2.342	1.780	2.405	1.780	2.483
29	1.812	1.842	2.404	1.842	2.467	1.842	2.545
30	1.875	1.905	2.467	1.905	2.530	1.905	2.608
31	1.938	1.968	2.530	1.968	2.593	1.968	2.671
32	2.000	2.030	2.592	2.030	2.655	2.030	2.733
33	2.062	2.092	2.654	2.092	2.717	2.092	2.795
34	2.125	2.155	2.717	2.155	2.780	2.155	2.858
35	2.188			2.218	2.843	2.218	2.921
36	2.250			2.280	2.905	2.280	2.983
37	2.312			2.342	2.967	2.342	3.045
38	2.375			2.405	3.030	2.405	3.108
39	2.438			2.468	3.093	2.468	3.171
40	2.500			2.530	3.155	2.530	3.233
41	2.562					2.592	3.295
42	2.625					2.655	3.358
43	2.688					2.718	3.421
44	2.750					2.780	3.483
45	2.812					2.842	3.545
46	2.875					2.905	3.608

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